Programming Lab 5 — Exception Handling

Computer Haven Seminar Registration System

Overview

Computer Haven is a small business that offers instructional computer seminars to local companies. The seminar is held in a conference room that accommodates up to 125 attendees.

Computer Haven needs a program that allows the sales manager to register multiple companies for an upcoming seminar, calculate registration fees, and generate a summary report. The program must also use exception handling to prevent invalid data from causing the program to crash.

This lab builds upon previous concepts including:

- variables and expressions
- branching
- loops
- accumulators
- input validation

and introduces:

- try/except
- multiple exception blocks
- raising exceptions

Reminder

Programming labs often require multiple attempts and revisions. Debugging and refining your program are normal parts of learning to program. Focus on building a working solution first, then improve and refine your code.

⭐ Challenge Extensions (Optional)

Challenge Extensions are optional enhancements for students who would like additional practice or a more advanced programming challenge. Students should focus on completing the required functionality first before attempting extensions.

Business Rules

The seminar room can hold a maximum of **125 attendees**.
The registration fee per person depends on the number of attendees registered by a company:

Number of Attendees	Cost Per Person
1–3	$150
4–9	$100
10 or more	$90

Program Requirements

The program should repeatedly:

1. Prompt for a company name.
2. Prompt for the number of attendees from that company.
3. Calculate and display the total registration cost for that company.
4. Ask if another company should be entered.

The program should continue until the sales manager chooses not to enter another company.

Validation Requirements

Attendee Count

The number of attendees must:

- be a whole number
- be at least 1
- not cause the seminar capacity to exceed 125 attendees

Seminar Capacity

If a registration would exceed the seminar capacity:

- Display “Seminar capacity exceeded”
- The registration should not be accepted.

Exception Handling Requirements

Requirement 1

Use a **try/except block** whenever the user enters the number of attendees.

Requirement 2

Implement at least **two exception blocks**.

Example:
except ValueError:
and
except Exception:

Requirement 3
Use **raise** to generate an exception when a business rule is violated.

Example:
if total_registered + attendees > 125:
raise ValueError("Seminar capacity exceeded.")

You may use this example or create your own solution.

Required Program Output

For each company display something like:
That will cost Microsoft $400.00

End-of-Program Summary
When all registrations have been entered, display:

Seminar Registration Summary

Companies Registered: 2
Total Attendees: 6
Total Revenue: $700.00
Average Revenue Per Attendee: $116.67

Program Must Include

- At least one while loop
- Input validation
- Accumulators
- Counters
- A try/except structure
- Multiple exception blocks
- At least one raise statement
- Meaningful comments
- Formatted output

Sample Run

Enter company name: Microsoft
How many from Microsoft will attend: 4

That will cost Microsoft $400.00

Another company? (y/n): y

Enter company name: Adobe
How many from Adobe will attend: 2
That will cost Adobe $300.00

Another company? (y/n): n

Seminar Registration Summary

Companies Registered: 2
Total Attendees: 6
Total Revenue: $700.00
Average Revenue Per Attendee: $116.67

⭐ Challenge Extensions (Optional)

Students seeking an additional challenge may add one or more of these extensions:

- Track the company with the largest registration
- Display the average company registration size
- Count how many companies qualify for each pricing tier
- Allow the user to cancel an individual registration
- Display remaining seminar capacity after each registration

Submission Requirements

Submit:

- .py file
- Test cases
- Code walk-through video (3–4 minutes)
